The third prefix at a singleton end

Human Compression project: L044 continuation

Retain the critical singleton hypotheses, the actual natural omission $G_0 = \mathrm{Alt}(\Omega \setminus \{a\})$, and the complete prefix-growth proof. Thus all marks after ρ_0 fix a, $C_k = a^{G_{\le k}}$ grows strictly, and distant later marks fix C_k pointwise. Actual ambient IP is assumed throughout.

Theorem 1. *If $P_2 = \langle \rho_0, \rho_1, \rho_2 \rangle$ is intransitive on its support, that support has at least eight points. Consequently a first shared label $g = 3$ cannot have an intransitive prefix in the critical singleton support interface.*

Proof. Suppose P_2 moves at most seven points. Pad this set with fixed points if necessary, using $n \ge 14$, and name $a = 1$. Every nonidentity even involution on seven points is a double transposition. Since ρ_0 moves a, we may normalize

$$\rho_0 = (1\ 2)(3\ 4).$$

The mark ρ_2 fixes 1 and commutes with ρ_0, so fixes 2 and preserves $\{3, 4\}$. On the other three points an involution has at most one transposition. Evenness and nonidentity therefore force, after renaming these points,

$$\rho_2 = (3\ 4)(5\ 6),$$

with 7 fixed. Strict growth $C_0 \subsetneq C_1$ makes ρ_1 move 2, while it fixes 1. Write

$$\rho_1 = (2\ x)(y\ z).$$

The independent interchanges of $3, 4$ and of $5, 6$ preserve both displayed marks, so we may take $x \in \{3, 5, 7\}$. The pair $\{y, z\}$ is any two-element subset of the remaining four points of $\{2, \dots, 7\}$. These are exactly the eighteen cases below; no group census is needed.

x	possible $\{y, z\}$	consequence
3	$\{4,5\}, \{4,6\}$	P_2 transitive on its support
3	$\{4,7\}, \{5,6\}, \{5,7\}, \{6,7\}$	$C_1 = C_2$
5	$\{3,6\}, \{4,6\}$	P_2 transitive on its support
5	$\{3,4\}$	preserved exterior pair
5	$\{3,7\}, \{4,7\}$	preserved exterior triple
5	$\{6,7\}$	actual point-stabilizer contradiction
7	all six possible pairs	$C_1 = C_2$

Here are the details of every row. In the first and third rows, the transpositions of the three marks join precisely $\{1, \dots, 6\}$, and 7 is fixed. Their action is therefore transitive on the entire support, contrary to the hypothesis. In the second row, $C_1 = C_2$ is $\{1, 2, 3, 4, 7\}$ for the pair $\{4, 7\}$ and is $\{1, 2, 3, 4\}$ for the other three pairs. In the last row, ρ_2 fixes $C_1 = \{1, 2, 7\}$ pointwise, again giving $C_1 = C_2$. Strict growth excludes all these cases.

For $x = 5$, $\{y, z\} = \{3, 4\}$, the two nontrivial P_2-orbits are

$$C_2 = \{1, 2, 5, 6\}, \qquad E = \{3, 4\}.$$

The P_1-orbits are $C_1 = \{1, 2, 5\}$, E, and fixed singletons. Thus ρ_3, which centralizes P_1, preserves the unique two-point orbit E. Every mark of index at least four centralizes P_2 and likewise

preserves E. The first three marks preserve it by its definition as a P_2-orbit. Hence E is a proper G-invariant set, contrary to transitivity.

For $x = 5$ and $\{y, z\} = \{3, 7\}$ or $\{4, 7\}$, the two nontrivial P_2-orbits are

$$C_2 = \{1, 2, 5, 6\}, \qquad E = \{3, 4, 7\}.$$

The two nontrivial P_1-orbits are $C_1 = \{1, 2, 5\}$ and E. The mark ρ_3 centralizes P_1 and fixes 1, so fixes C_1 pointwise. It therefore cannot exchange these two three-point orbits, and must preserve E. All marks of index at least four centralize P_2, whose nontrivial orbits have distinct sizes four and three, so also preserve E. Once again G preserves E, a contradiction.

Finally suppose $\rho_1 = (2\ 5)(6\ 7)$. On $\{1, 2, 5\}$ the product $\rho_0\rho_1$ is a three-cycle, and off this set it is $(3\ 4)(6\ 7)$. Thus

$$w = (\rho_0\rho_1)^3 = (3\ 4)(6\ 7).$$

This is a nonidentity even permutation fixing $a = 1$, so belongs to the actual natural group G_0. It also belongs to P_1. Full IP gives

$$w \in P_1 \cap G_0 = \langle \rho_1 \rangle.$$

But w fixes 2, whereas ρ_1 moves it. This is impossible. All eighteen choices are exhausted.

The first-shared support theorem gives $|\operatorname{supp}(P_{g-1})| \leq 2g + 1$. At $g = 3$ its intransitive-prefix alternative would give the forbidden support bound seven, proving the consequence. □

Attribution and boundary. The complete singleton-prefix growth and support proofs are retained. The finite normalization here is a direct hand argument on three literal permutations; an exploratory enumeration suggested its case split but is not a proof premise. The point-stabilizer step uses the acquired actual natural omission and full IP, and does not assign IP or fidelity to any orbit projection. The source route is the singleton continuation of Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, Propositions 5.28–5.32. Larger first shared indices and complete singleton routing remain. No whole-original, question, numerical or formal credit is claimed.